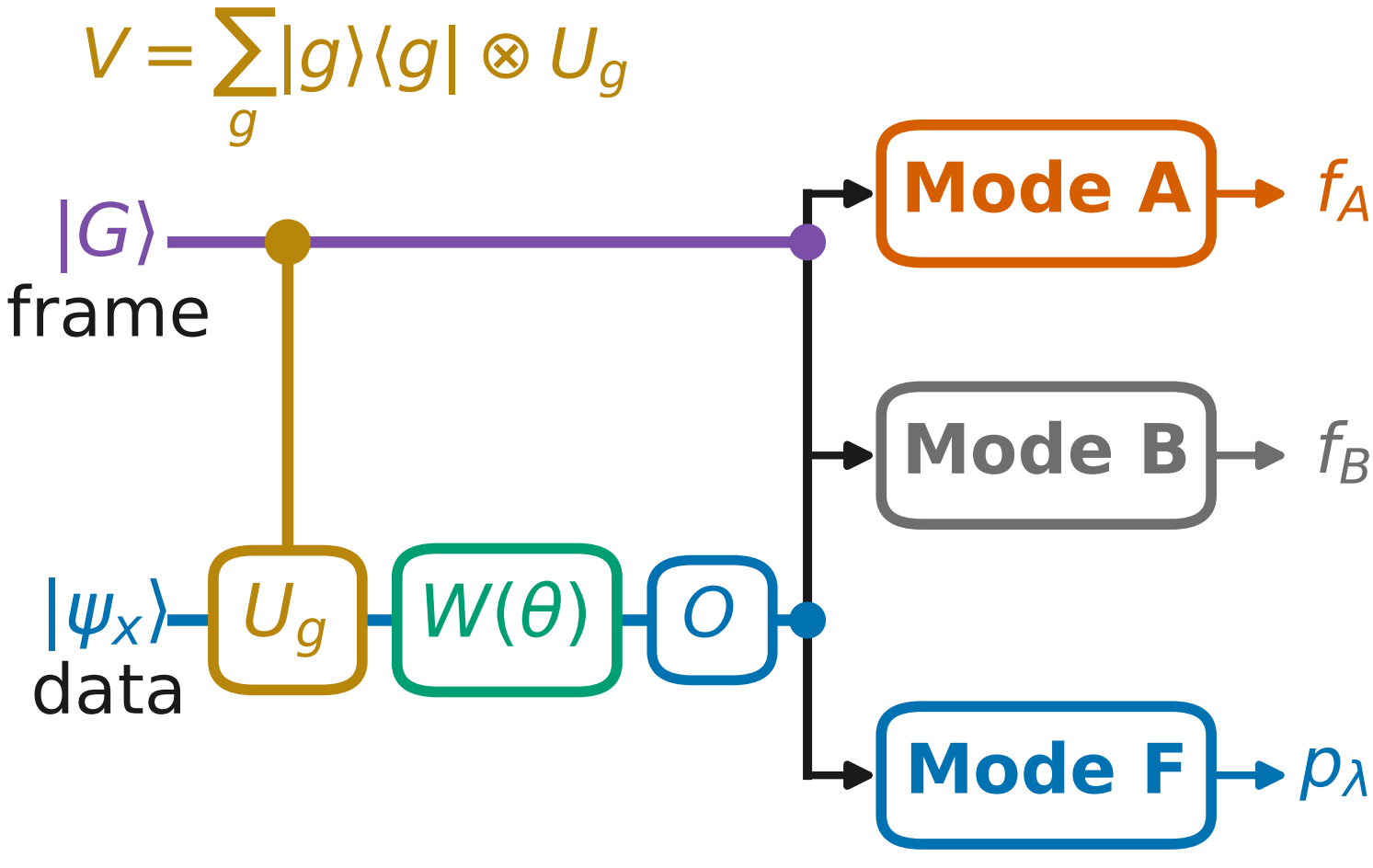

$$\Pi = \frac{1}{|G|}\sum_g U_g, \quad H_+ = \text{range}(\Pi)$$